

What it does `useLayerDetailGraph` builds the node and edge data that React-Flow renders for the layer-detail view. It pulls state from the dashboard store (selected node, search results, diff mode, etc.), computes expanded child nodes, container overlays, and then merges all this into a final `nodes` and `edges` array that includes visual flags for search hits, selection, tours, and diff status.

How it works1. State extraction – several `useDashboardStore` calls pull IDs and flags.2. **Expanded children*

- `useMemo` iterates over `expandedContainers`, looks up cached child positions, and creates `Node` objects with `parentId: containerId` and `extent: "parent"`.3. **Overlay maps*
- `searchHitsByContainer`, `diffContainers`, `focusContainerIds`, and `selectionContainerIds` are memoized sets/maps that map container IDs to hit counts or flags.4. **Node composition*
- `nodes` is a memo that concatenates `topo.nodes` and `expandedChildNodes`. For each node it:
 - Skips portal nodes.
 - For container nodes, updates `data` with `isExpanded`, `hasSearchHits`, `searchHitCount`, `isDiffAffected`, and `isFocusedViaChild` only if any flag changed.
 - For regular nodes, sets `isHighlighted`, `searchScore`, `isSelected`, `isTourHighlighted`, `isNeighbor`, and `isSelectionFaded` based on the current selection and search results, again only creating a new object when a visual flag changes.
- 5. **Edge inflation*
- `expandedEdges` replaces aggregated container-to-container edges with the underlying file-to-file edges from `topo.filteredEdges`. It also injects intra-container edges for expanded containers, deduplicating by a composite key.6. **Final edges*
- `edges` merges `expandedEdges` with `topo.portalEdges` and applies selection styling if a node is selected.

Notes for reviewers- The hook relies heavily on memoization; verify that dependencies cover all used values.- The `expandedChildNodes` logic assumes `containerLayoutCache` contains `childPositions`; if the cache is missing a child, that child is silently omitted.- Edge inflation uses a `seen` set keyed by `"src|tgt|type"`; ensure this key is unique enough for all edge types.- The selection-based edge styling section is incomplete in the snippet (`edge.source === selectedNo`); confirm the final implementation matches the intended logic.